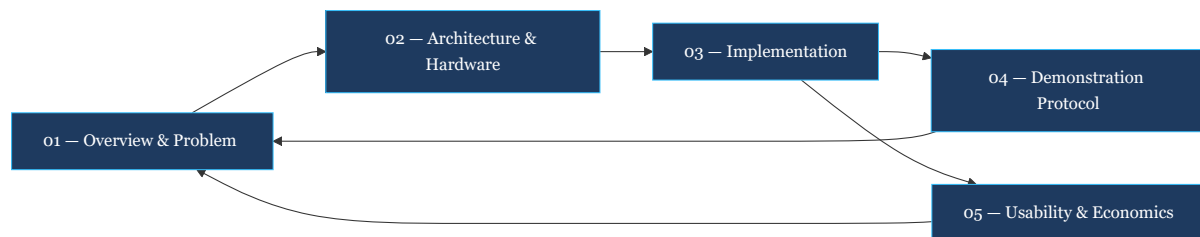


PI-CLOUD — Annual Day Project Index

PROJECT SUMMARY

Self-hosted private cloud on a Raspberry Pi 5, replacing consumer cloud subscriptions. **Capital: \$186 / ₹16,500. Recurring: ≈\$6 / ₹450 per year.**
CasaOS + Nextcloud (per-user encryption) + Jellyfin + Samba, entirely on the local network, with no inbound internet exposure.

Documentation set



#	Document	Answers	Primary evidence
01	01-Overview-and-Problem	Why the project exists; requirements; risks	§8 criteria, §9 risk register
02	02-Architecture-and-Hardware	What it is; hardware, software, network, security model	Measured throughput, memory budget, port map
03	03-Step-by-Step-Implementation	How it was built; whether the claims are verified	30-point checklist , restore test, DR runbook
04	04-Interactive-Demo-Guide	How the claims are demonstrated and assessed	Demonstration sequence, evidence artefacts, verification sheet
05	05-Everyday-Usability	Is it worth deploying for a real household	TCO + sensitivity, functional trade-offs, adoption analysis

Headline figures

Metric	Value	Verified by
Capital cost	\$186 / ₹16,500	03 §5, 02 §2
Recurring cost	≈\$6 / \$0.50 per month	02 §12
Break-even	≈2.8 months (full stack) / ≈10 months (storage only)	05 §2.4

Metric	Value	Verified by
Build time, critical path	≈45 minutes	03 Summary
OS failure recovery	≈25 min, no data loss	03 §14 DR-2
Power	≈6 W average, ≈53 kWh/year	02 §12
Concurrent users	6 mixed workloads	04 D-4
Concurrent 1080p streams	3–4	04 D-4
Storage redundancy	None — single disk	02 §11
Inbound internet exposure	None	02 §6.2, §10

Where to start

If you want to...	Go to
Understand the reasoning	01-Overview-and-Problem
Review the technical design	02-Architecture-and-Hardware
Reproduce the build	03-Step-by-Step-Implementation
Assess the system	[[04-Interactive-Demo-Guide#7. Independent verification sheet
Check the economics	[[05-Everyday-Usability#2. Total cost of ownership
See the failures and limits	01 §9 and 04 §10

READ THE LIMITATIONS

[01 §9](#) and

[04 §10](#) document

10 known limitations, of which the two material ones are **single-disk storage** and **no uninterruptible power supply**. Both are mitigated for ≈\$95 total; see [05 §10](#).

The project is explicit about what it does not do. That is a deliberate property of the assessment, not an omission from it.